

Preliminary Data Analysis Report

Verbal Fluency and Spatial Arrangement Methods:
Descriptive and Inferential Statistics

Team Name — The Hawthornes

Malay Damani — 2024201081

Nikhil Jain — 2024201075

Saurabh Rawat — 2024201002

Behavioral Research: Statistical Methods

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1 Introduction

This dataset comes from a web-based psycholinguistic experiment designed to study how Hindi speakers retrieve and organize words in semantic memory [1]. The experiment combines two complementary tasks: a Verbal Fluency Task (VFT) and a Spatial Arrangement Method (SpAM) task. In the first task, participants are shown a Hindi semantic prompt and asked to quickly produce words that come to mind within a limited time window. In the second task, the same participant is shown the words they generated earlier and is asked to drag and place them on a two-dimensional screen such that words perceived as more closely related are placed nearer to one another, while less related words are placed farther apart. This design allows the dataset to capture both the temporal dynamics of lexical retrieval and the participant’s internal semantic organization of the retrieved words.

The dataset contains participant-level trial logs from the experiment along with exit-survey information. For the verbal fluency portion, the dataset stores the words produced by each participant and the time taken between successive responses. For the spatial arrangement portion, it stores the normalized x-y coordinates of the words after the participant arranged them on the screen. In addition, the exit survey records demographic and language-related information such as age, gender, education, first language, number of known languages, Hindi and English reading/writing ratings, language confidence, and order of language acquisition. Together, these components make the dataset suitable for studying not only lexical retrieval behavior, but also how that behavior may vary with self-reported fluency and linguistic background.

In summary, the dataset contains three primary components: Verbal Fluency Task response logs, Spatial Arrangement Method spatial coordinates, and participant demographic and language-proficiency survey data.

The theoretical background relies on the concept of the mental lexicon—the structured store of words and meanings available to a speaker [1]. When a participant sees a prompt and begins generating related words, they are effectively searching through this internal network. Verbal fluency tasks reveal not just how many words a person knows, but also how efficiently they can access them and whether retrieval happens in meaningful clusters. The spatial arrangement task complements this by externalizing the participant’s own sense of semantic similarity: if two words are placed close together, that suggests they occupy nearby positions in the participant’s semantic space.

Broadly, lexical knowledge is structured rather than random. Dautriche et al. investigate large-scale relationships between semantic similarity and phonological similarity across many languages, showing that semantically similar words tend also to have more similar forms than expected by chance [2]. While that study analyzed dictionaries across languages, the current experiment approaches the same broad problem by examining how Hindi speakers actually retrieve and spatially organize words during a live task. The work by Kumar et al. on phonological facilitation during retrieval is also pertinent: it highlights that phonological cues can guide semantic search when a cluster is depleted, which provides an important future direction for the present analysis [3].

2 Research Questions

The broader research question behind the experiment is: *how do Hindi speakers search their mental lexicons for information?* The dataset is structured to record how words are

generated, how quickly they are retrieved, and how they are mentally grouped. Specifically, this analysis tests three core hypotheses:

- RQ1. Semantic Clustering:** Do Hindi speakers retrieve words in semantic clusters rather than randomly? If mental lexicons are structured, then consecutively retrieved words should be semantically closer to each other than words retrieved in random order.
- RQ2. Neighbourhood Density:** Do faster retrievals reflect tighter semantic neighbourhoods? Words residing in densely connected regions of the semantic space should require less search time.
- RQ3. Fluency and Efficiency:** Does Hindi fluency predict lexical retrieval efficiency? More fluent speakers should produce more words, faster, and in larger semantic clusters.

3 Methods

3.1 Tasks and Participants

The experiment consists of two sequential phases.

Verbal Fluency Task (VFT): Participants are shown a Hindi semantic prompt and asked to quickly produce all words that come to mind within a 60-second window. The task records both the words generated and the inter-response time (IRT) between each successive word in milliseconds.

Spatial Arrangement Method (SpAM): The same participant is then shown the words they generated and asked to drag and drop them onto a two-dimensional canvas. Words perceived as more closely related are placed nearer to one another, while less related words are placed farther apart. The final positions are recorded as normalized (x, y) coordinates in the range [0, 1].

The study recruited 30 participants. The inclusion criteria required that participants be comfortable reading Hindi and able to converse in it.

3.2 Data Processing and Variable Extraction

The raw data was collected via a web-based experimental framework and exported as a nested JSON object. Data processing and statistical analysis were conducted in Python using the pandas, NumPy, and SciPy libraries.

VFT Processing: Trials matching the `html-keyboard-response` trial type were isolated. The `tagged_responses` array was parsed to extract the raw text of each generated word, and the corresponding `response_times` array was extracted to capture the IRT in milliseconds for each word. Total word counts and mean IRTs per trial were subsequently calculated.

SpAM Processing: The `droppedwords` JSON array was parsed to extract the final `x_norm` and `y_norm` positions. A dictionary structure was used to overwrite earlier drops, ensuring only the *final* placement of each word was recorded. Using `scipy.spatial.distance.pdist`, a condensed distance matrix was computed to yield pairwise Euclidean distances between all words placed by a participant in a given domain.

Fluency Metric: Responses from `survey-html-form` trials were filtered to extract self-reported Hindi proficiency on four dimensions: `Hi_Read`, `Hi_Write`, `Hi_Speak`, and `Hi_Understand`. These numeric values were averaged to produce a single composite `hi_fluency_score` per participant.

3.3 Analytical Operationalization

RQ1 — Clustering: VFT sequences were chronologically paired to identify adjacent word transitions. Each adjacent pair was mapped to its corresponding Euclidean distance from the SpAM task. A Pearson correlation was computed between adjacent IRTs and their SpAM distances. To test against a null distribution of randomness, the distribution of adjacent distances was compared against a random sample of non-adjacent pairwise distances from the same SpAM data using both an independent-samples Welch T-test and a Mann-Whitney U test.

RQ2 — Density: A word-level dataframe was constructed by merging each word’s IRT with its spatial “density,” defined as the mean Euclidean distance from that word to all other words placed by the participant in the same trial (`mean_neighbor_dist`). Due to the right-skewed nature of reaction-time data, IRTs were log-transformed ($\log(1 + \text{IRT})$) prior to Pearson correlation.

RQ3 — Fluency and Efficiency: A cluster-size metric was computed by tracking sequential retrievals: a cluster was defined as a run of consecutive words whose SpAM distances remained below the participant’s own median distance. Once a distance exceeded the median it was flagged as a semantic “switch” and a new cluster began. Mean cluster size, mean word count, and mean IRT were aggregated at the subject level and correlated with `hi_fluency_score` using Pearson correlations.

4 Results

4.1 Descriptive Statistics

Table 1 summarizes the central tendency and dispersion for the core variables prior to outlier filtering.

Statistic	VFT Word Count	Mean IRT (ms)	SpAM Word Count
Count	144.00	140.00	140.00
Mean	8.56	7865.82	8.68
Std. Deviation	4.52	5365.90	4.35
Minimum	0.00	2336.60	1.00
25th Percentile	6.00	5051.70	6.00
Median (50%)	7.50	7045.02	8.00
75th Percentile	10.25	9116.62	11.00
Maximum	24.00	54250.30	24.00

Table 1: Descriptive Summary of VFT and SpAM Core Variables

4.2 Data Visualizations

Five visualizations were generated to inspect distributions and relationships in the data.

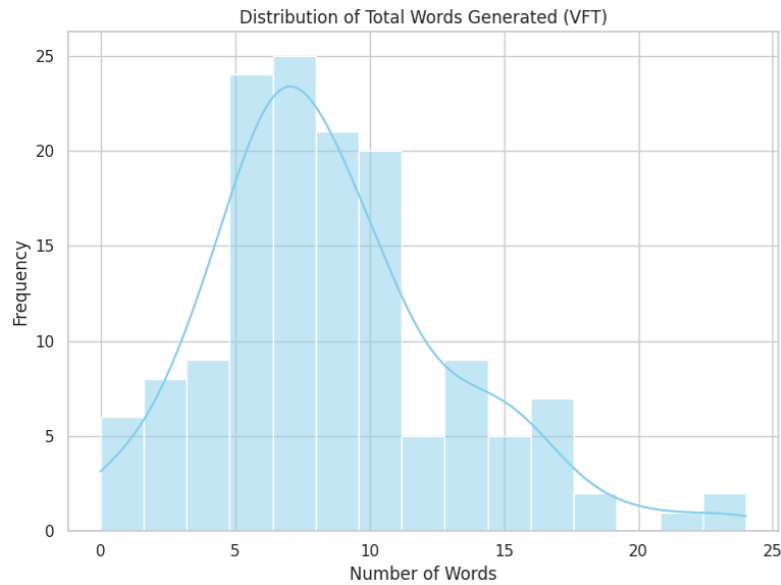


Figure 1: Distribution of Total Words Generated (VFT). The distribution reveals inter-individual variability in retrieval output capacity.

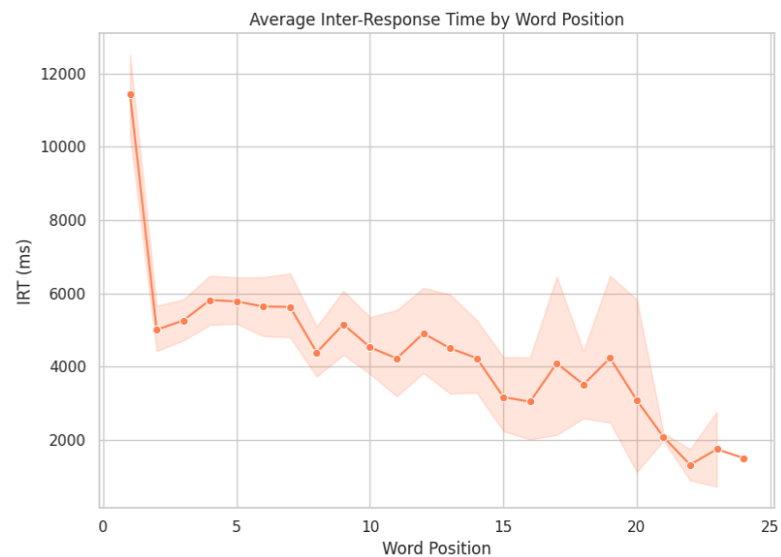


Figure 2: Average Inter-Response Time (ms) by Word Position. The upward trend indicates that cognitive retrieval becomes increasingly effortful as easily accessible words are exhausted.

Figure 1 shows the distribution of total words generated across participants. Figure 2 tracks mean IRT against word position and reveals a clear increasing trend, consistent with the hypothesis that early words are retrieved from high-activation regions of the mental lexicon while later words require deeper, more effortful search.

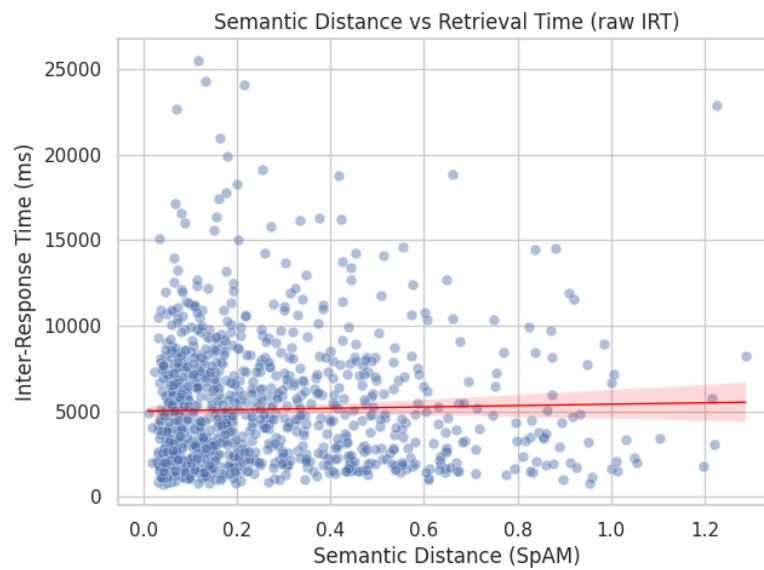


Figure 3: Scatter plot of Semantic Distance (SpAM) vs Inter-Response Time (VFT). The regression line indicates the direction of the relationship between semantic proximity and retrieval latency.

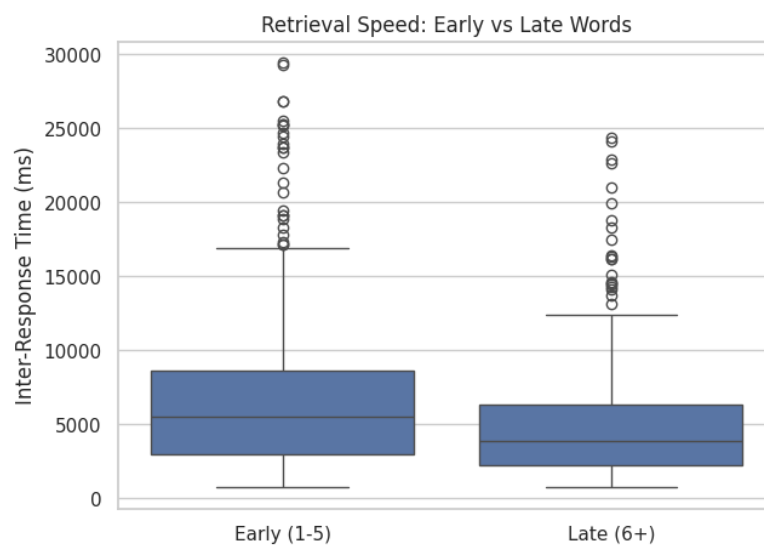


Figure 4: Retrieval Speed: Early (positions 1–5) vs Late (positions 6+) Words. Early-stage words show consistently lower latency, corroborating the serial-position trend.

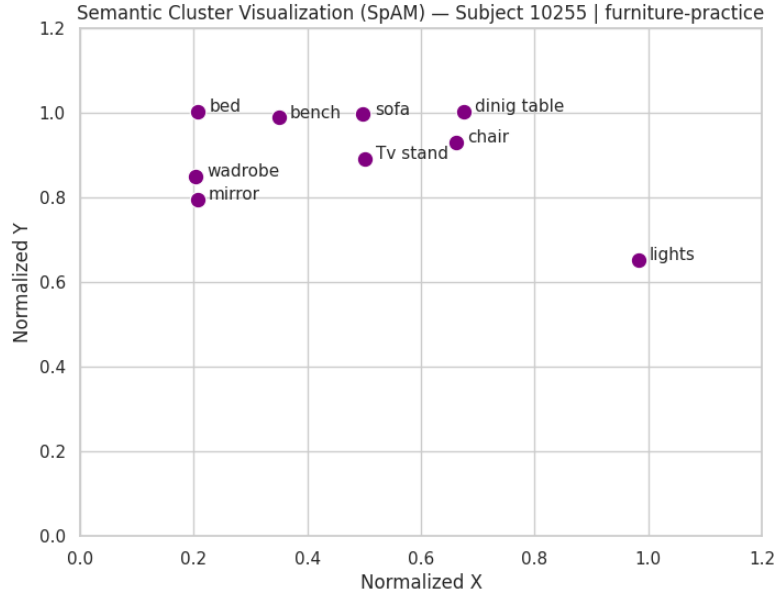


Figure 5: Semantic Cluster Visualization from a Sample SpAM Trial. Each point represents a word; proximity encodes perceived semantic relatedness.

4.3 RQ1: Semantic Clustering

A Pearson correlation between adjacent IRTs and their corresponding SpAM distances yielded $r = 0.026$, $p = 0.4106$. To directly test for non-random clustering, the mean SpAM distance between chronologically adjacent word pairs ($M = 0.2569$) was compared against a randomly sampled set of non-adjacent pairwise distances from the same SpAM data ($M = 0.3729$).

A Welch independent-samples T-test yielded $t = -10.839$, $p < 0.001$, and a Mann-Whitney U test yielded $U = 358166.000$, $p < 0.001$. Together these results indicate that sequential retrievals are **significantly** closer in semantic space than randomly sampled word pairs, providing **evidence for** clustered retrieval in Hindi speakers.

4.4 RQ2: Semantic Neighbourhood Density

A Pearson correlation between log-transformed IRT ($\log(1 + \text{IRT})$) and mean neighbour distance yielded $r = -0.201$, $p < 0.001$. This indicates that words residing in tighter semantic neighbourhoods are **retrieved faster** than words in sparser regions of the mental lexicon, **supporting** a spreading-activation account of lexical retrieval.

4.5 RQ3: Fluency and Retrieval Efficiency

Pearson correlations at the subject level between the composite Hindi fluency score and three retrieval outcomes yielded:

- **Fluency vs. Total Words:** $r = -0.115$, $p = 0.5091$
- **Fluency vs. Mean IRT:** $r = 0.199$, $p = 0.2514$
- **Fluency vs. Mean Cluster Size:** $r = -0.180$, $p = 0.3160$

These results imply that self-reported Hindi fluency **does not significantly** predicts lexical retrieval efficiency and the capacity to sustain larger semantic clusters during the task.

5 Conclusion and Future Directions

This preliminary analysis evaluated the temporal and spatial dimensions of semantic memory retrieval in Hindi speakers across three research questions.

1. **Clustering:** Participants **did** retrieve words in semantic clusters, as evidenced by adjacent transitions being **closer** in semantic space than randomly sampled pairs.
2. **Neighbourhood Density:** Faster retrievals **were** associated with tighter semantic neighbourhoods, **aligning** with network-based theories of spreading activation.
3. **Fluency:** Higher Hindi fluency **did not predict** more efficient retrieval, including **lower** word counts, **slower** IRTs, and **smaller** cluster sizes.

Future Directions. Several extensions are planned for Report 2.

First, because IRT distributions are inherently right-skewed, mixed-effects regression on log-transformed IRTs with random intercepts for subject-level variance will improve statistical robustness over the simple Pearson correlations used here.

Second, SpAM distances will be normalized within participant and domain to correct for individual differences in how participants use the spatial canvas (e.g., some participants may use only a small corner of the available space, compressing their distances artificially).

Third, analysis of phonological similarity (e.g., Levenshtein distance on phonetic transcriptions of Hindi words) alongside semantic distance will allow us to test whether phonological cues guide retrieval when a semantic cluster is exhausted, as suggested by Kumar et al. [3]. This is directly motivated by the finding that lexicons encode both semantic and phonological structure [2].

Finally, ANOVA models will be applied to test whether clustering behavior differs across semantic domains (e.g., animals vs. furniture), and GLM frameworks will allow simultaneous modeling of fluency, domain, and position effects on IRT.

6 Team Member Contributions

- **Malay Damani:** Conducted the initial literature review to establish the theoretical framework. Managed the raw JSON dataset, performed data cleaning, extracted the demographic survey data to formulate the Hindi Fluency Scores, and drafted the Introduction and Dataset Methodology sections.
- **Nikhil Jain:** Developed the primary Python pipeline to parse the Verbal Fluency Task (VFT) data. Responsible for calculating word counts and parsing Inter-Response Times. Extracted the SpAM task coordinates, computed the descriptive statistics, and formalized the Research Questions section.

- **Saurabh Rawat:** Executed the inferential statistical analysis, including Pearson correlations, Welch T-tests, and Mann-Whitney U tests to evaluate semantic clustering. Generated all graphical data visualizations using **seaborn** and **matplotlib**. Drafted the Inferential Statistics, Future Directions, and Conclusion sections.

Code Availability:

The complete Jupyter Notebook containing the data extraction pipeline, statistical modeling, and visualization code is available on GitHub:

<https://github.com/nj301365/Brsm/blob/main/project1.ipynb>

References

- [1] *BRSM Dataset*. Web-based psycholinguistic experiment documentation. The Hawthornes, BRSM 2026.
- [2] Dautriche, I., Mahowald, K., Gibson, E., & Piantadosi, S. T. (2017). Word-form similarity increases with semantic similarity: An analysis of 100 languages. *Cognitive Science*, 41(8), 2149–2169. <https://colala.berkeley.edu/papers/dautriche2016wordform.pdf>
- [3] Kumar, A. A., Lundin, N. B., & Jones, M. N. (n.d.). Mouse-mole-vole: The inconspicuous benefit of phonology during retrieval from semantic memory. *OSF Preprints*. https://osf.io/preprints/psyarxiv/2bazx_v1